

MGMT 675: Generative AI for Finance

FMA 2025

Kerry Back

Jones Graduate School of Business

Rice University

FMA IITA Session: October 24, 2025

- Half-semester MBA course at end of 1st year
- Course comes after:
 - Core finance (semester)
 - Excel-based Applied Finance (half-semester)
- Finance topics mostly repeated from prior courses
- Can be adapted to other courses of different lengths, at different points in the curriculum, and for different student groups

Derek Waldron, JP Morgan Chief Analytics Officer:

- (What we're working towards is that) every employee will have their own personalized AI assistant; every process is powered by AI agents, and every client experience has an AI concierge.
- You'll still have people at the top who are managing and have relationships with clients, but many, many of the processes underneath are now being done by AI systems.

Workers would shift from being creators of reports ... or "makers" ... to "checkers" or managers of AI agents doing that work.

- As AI models begin to handle underwriting, compliance, and asset allocation, the traditional architecture of financial work is undergoing a fundamental shift.
- As job descriptions evolve, so does the definition of financial talent. Excel is no longer a differentiator. Python is fast becoming the new Excel.
- But technical skills alone will not cut it. The most in demand profiles today are those that speak both AI and finance.

What Should We Teach about AI?

- Where is it going?
 - Point-and-click GUIs will be replaced by agentic chatbots.
 - OpenAI this month introduced OpenAI apps. Launched with

What to do about AI?

It should be used in every class, and it will be, but it's not yet. So, a course specifically on AI has a role to play.

What to do in a course specifically on AI for finance?

- Teach students about AI tools and how they are used in industry (custom chatbots, agents, RAG, ...)
- Give students experience in collaborating with AI: financial analysis and building things useful for financial analysis

Principles (Should Remain Constant)

1. Working with AI is a dialogue: prompt, evaluate, iterate
2. AI doesn't do numbers, so it needs to generate code
3. AI is just a tool, and domain knowledge remains important

Principles (Should Remain Constant)

1. Working with AI is a dialogue: prompt, evaluate, iterate
2. AI doesn't do numbers, so it needs to generate code
3. AI is just a tool, and domain knowledge remains important
4. Domain knowledge + AI + code $\Rightarrow$ ★Superstar★

Tools (Definitely Not Constant!)

- In the past, I used Julius.ai, which is an easy-to-use chatbot with a 'Code Interpreter' connected to Anthropic or OpenAI models.
- But, Anthropic's MCP (11-25-2024) and now Claude Skills (10-16-2025) have changed the game.
- Skills are customized prompts. Claude only reads them when needed, so you can have many without overloading the context window.

Claude Code or Claude Desktop?

- As AI has improved in general, the advantage of a chatbot that can execute terminal commands over one that cannot has grown.
- With Claude skills, Claude Code is a general agent.
- So, I think it is time to use a professional setup: Python, Claude Code, and VS Code. Install Claude Code. Let Claude Code install everything else.
- If not that, then Claude Desktop. But, best: Claude Desktop for every finance course, Claude Code for AI in Finance course.

Claude Example

First, install Excel and Word skills (<https://github.com/anthropics/skills>). Ask Claude to do it after researching how (just downloads text files to the right folders). Only has to be done once.

Claude prompt (after uploading *Valuing Wal-Mart 2010*):

Read the case but ignore the valuation method described in it. Do a 2-stage DCF analysis. Document your assumptions and explain your reasons for them. Create professionally formatted Excel and Word docs.

Creating Apps: Basic Example

Claude prompt:

Create a streamlit app that prompts a user to input their name and then displays a window with Hello <name>! centered in a large font accompanied by a display of balloons. Run the app and tunnel it using my ngrok key stored at C:\users\kerry\.env.

An Idea for How to Assess

Group projects:

- Financial analysis (submit chat, product, and written assessment). Defend assignments orally.
- Build app (submit app and written description and evaluation) – could be Claude skill instead
- Build chatbot (submit chatbot and written description and evaluation) – could be Claude skill instead

In-class seated exam: Analyze a case and prepare Excel/Word/... reports **using AI**.

Website to Support Teaching Gen AI for Finance

genai4finance.kerryback.com

- Course description aimed at teachers
- Blog: short posts about teaching various topics on AI and finance
- Course materials: 2025 and 2026 (partial)
- Slides (these and upcoming talk)
- Python materials (for pre-course workshop or individual study)

